Drainage

The big idea

Drainage describes the river system of an area. Water from rain and melting snow flows downhill into streams and rivers, which together form a network. India's rivers fall into two big groups, the **Himalayan** and the **Peninsular** rivers, which behave very differently.

KEY TERMS

Drainage basin: the area drained by a single river system. **Water divide:** an upland (mountain or high ground) that separates two neighbouring drainage basins; for example, the **Ambala** area lies on the divide between the Indus and Ganga systems. **Perennial river:** flows all year. **Seasonal river:** flows mainly in the rainy season.

Drainage patterns

- **Dendritic**: like the branches of a tree; forms where the rock is uniform (e.g. rivers of the northern plains).
- **Trellis**: tributaries join the main river at nearly right angles where hard and soft rocks lie parallel.
- **Rectangular**: develops on strongly jointed rocky terrain.
- **Radial**: streams flow in all directions from a central peak or dome (e.g. **Amarkantak** range).

The Himalayan rivers

The major Himalayan rivers are the **Indus, Ganga and Brahmaputra**. They are **perennial**, fed by both rain and **melted snow** from glaciers. They carve deep **gorges**, carry huge loads of silt and sand, form **meanders, oxbow lakes**, flood plains and **deltas** in their lower courses.

The Indus system

- Rises in **Tibet near Lake Mansarovar**. Flows west, enters India in **Ladakh** (Jammu and Kashmir/Ladakh), forming a picturesque gorge.
- Tributaries in Kashmir: **Zaskar, Nubra, Shyok and Hunza**. From the Punjab region: **Satluj, Beas, Ravi, Chenab and Jhelum** join together near **Mithankot** in Pakistan.
- Length: **2,900 km**; one of the longest rivers in the world. Only a third of the basin lies in India (J&K, Himachal, Punjab); the rest in Pakistan.
- **Indus Water Treaty (1960):** India can use only **20%** of the total water carried by the Indus system.

The Ganga system

- Headwaters called the **Bhagirathi** are fed by the **Gangotri Glacier**; it joins the **Alaknanda** at **Devprayag** in Uttarakhand. The Ganga emerges onto the plains at **Haridwar**.
- Tributaries: **Yamuna** (rises from the Yamunotri Glacier, joins at **Prayagraj** (Allahabad)), Ghaghara, Gandak, Kosi (from Nepal; the Kosi is called the "sorrow of Bihar" for its floods). From the Peninsular uplands: **Chambal, Betwa and Son**.
- At **Farakka** in West Bengal it bifurcates: the **Bhagirathi–Hooghly** flows south through the deltaic plains to the Bay of Bengal; the main stream flows into **Bangladesh** and is joined by the Brahmaputra; further down it is called the **Meghna**.
- Length: over **2,500 km**. The **Ambala** is on the water divide between the Indus and Ganga systems.
- The **Sundarban Delta**, the world's **largest and fastest growing delta**, is formed by the Ganga and Brahmaputra; named after the **Sundari** tree that

grows in marshland; home to the Royal Bengal tiger.

The Brahmaputra system

- Rises in **Tibet** east of Mansarovar lake, very close to the sources of the Indus and Satluj. Slightly longer than the Indus; most of its course lies outside India.
- Flows east parallel to the Himalayas in Tibet, where it is called **Tsang Po**. Reaching **Namcha Barwa** (7,757 m), it takes a U-turn and enters **Arunachal Pradesh** through a gorge. Here it is called the **Dihang**. It is joined by the **Dibang, Lohit** and many tributaries to form the Brahmaputra in **Assam**.
- In Tibet it carries less water and silt (cold, dry area); in India it carries a large volume of water and silt (heavy rainfall). It forms a **braided channel** with many riverine islands, including **Majuli**, the largest inhabited riverine island in the world. Floods in Assam and Bangladesh are common.

The Peninsular rivers

The main water divide in Peninsular India is formed by the **Western Ghats**. Most rivers flow **east** into the Bay of Bengal and form **deltas** (Mahanadi, Godavari, Krishna, Kaveri). Only the **Narmada and Tapi** flow **west** and make **estuaries**. Peninsular rivers are mostly **seasonal**, with shorter and shallower courses.

River	Source	Key facts
Narmada	**Amarkantak hills**, Madhya Pradesh	Flows west in a **rift valley** formed by faulting; creates the **Marble Rocks** near Jabalpur and the **Dhuadhar falls**; tributaries are short and join at right angles; MP and Gujarat
Tapi	**Satpura ranges**, Betul district (MP)	Flows west in a rift valley parallel to the Narmada; MP, Gujarat and Maharashtra
Godavari	Slopes of Western Ghats, **Nasik** district (Maharashtra)	**Largest Peninsular river**, about **1,500 km**; called **Dakshin Ganga**; tributaries Purna, Wardha, Pranhita, Manjra, Wainganga, Penganga

River	Source	Key facts
Mahanadi	Highlands of **Chhattisgarh**	About **860 km**; flows through Odisha to the Bay of Bengal
Krishna	Spring near **Mahabaleshwar**	About **1,400 km**; tributaries Tungabhadra, Koyana, Ghatprabha, Musi, Bhima; Maharashtra, Karnataka, Andhra Pradesh
Kaveri	**Brahmagiri** range of Western Ghats	About **760 km**; tributaries Amravati, Bhavani, Hemavati, Kabini; **Shivasamudram** falls (second biggest waterfall in India), its power supplies Mysore, Bengaluru and the Kolar Gold Field; Tamil Nadu and Karnataka

Smaller west-flowing rivers include the Sabarmati, Mahi, Bharathpuzha and Periyar.

Lakes

India has many lakes, differing in size and character.

- **Permanent** vs **seasonal** (in inland drainage basins of semi-arid regions).
- **Oxbow lakes**: formed when a meandering river is cut off.
- **Lagoons**: formed by spits and bars in coastal areas, e.g. **Chilika** (Odisha), **Pulicat** (Andhra Pradesh–Tamil Nadu), **Kolleru** (Andhra Pradesh).
- **Salt lakes**: in inland drainage, e.g. **Sambhar Lake** in Rajasthan; its water is used for producing salt.
- **Freshwater lakes**: most are in the Himalayan region, mainly of **glacial** origin. **Wular Lake** (J&K) is the **largest freshwater lake**, formed by **tectonic activity**. Others: **Dal Lake, Bhimtal, Nainital, Loktak** and **Barapani**.
- **Man-made lakes**: created by damming rivers for hydel power, e.g. **Guru Gobind Sagar** (Bhakra Nangal project).

Importance of lakes: regulate river flow and prevent floods; maintain an even flow in dry seasons; generate hydel power; moderate climate; maintain aquatic ecosystems; enhance natural beauty; promote tourism and recreation.

Role of rivers in the economy and river pollution

Rivers have been central to human history: settlements grew on their banks. River water is used for **irrigation, navigation, hydro-power generation**, drinking water and fishing. In India, an agricultural country, rivers are especially important.

River pollution: growing domestic, municipal, industrial and agricultural demand affects water quality. Untreated sewage and industrial waste are dumped into rivers. The **National River Conservation Plan (NRCP)** and the **Namami Gange** programme aim to clean rivers.

DELTA VS ESTUARY

Narmada and Tapi flow into the Arabian Sea through **rift valleys** and form **estuaries**, not deltas, because they carry little silt and flow through hard rock with a steep gradient. East-flowing rivers form **deltas**.

MAP POINTS

Rivers: Indus, Ganga, Yamuna, Brahmaputra, Narmada, Tapi, Godavari, Krishna, Kaveri, Mahanadi. Lakes: Wular, Dal, Sambhar, Chilika, Pulicat, Kolleru, Loktak.

Quick recap

- Himalayan rivers: perennial, snow-fed, gorges, meanders, deltas.
- Indus (2,900 km; Indus Water Treaty 1960, India uses 20%); Ganga (Bhagirathi + Alaknanda at Devprayag; Farakka); Brahmaputra (Tsang Po, Dihang, Majuli).
- Peninsular rivers: seasonal; Western Ghats is the main water divide; Narmada and Tapi flow west through rift valleys.

- Godavari is the largest Peninsular river (Dakshin Ganga).
- Lakes: Wular (largest freshwater), Chilika (largest saltwater lagoon), Sambhar (salt).

Practice questions

1. The Bhagirathi and the Alaknanda meet at
   (a) Devprayag (b) Haridwar (c) Prayagraj (d) Farakka
2. Which river is known as the Dakshin Ganga?
   (a) Krishna (b) Kaveri (c) Godavari (d) Mahanadi
3. According to the Indus Water Treaty (1960), India can use what share of the Indus system's water?
   (a) 50% (b) 10% (c) 33% (d) 20%
4. In Tibet, the Brahmaputra is known as the
   (a) Dihang (b) Tsang Po (c) Meghna (d) Lohit
5. Which of these rivers flows through a rift valley into the Arabian Sea?
   (a) Krishna (b) Mahanadi (c) Narmada (d) Kaveri
6. The largest freshwater lake in India is
   (a) Dal Lake (b) Chilika Lake (c) Sambhar Lake (d) Wular Lake
7. A drainage pattern in which streams flow in all directions from a central peak is called
   (a) dendritic (b) trellis (c) rectangular (d) radial
8. Distinguish between the Himalayan rivers and the Peninsular rivers.
9. What is the importance of lakes? Classify lakes with examples.

Answers: 1. a 2. c 3. d 4. b 5. c 6. d 7. d 8. see notes 9. see notes